38 There is even a proposal, from Lisa Randall, of Harvard, and Raman Sundrum, of Johns Hopkins, in which gravity too can be trapped, not by a sticky brane, but by extra dimensions that curve in just the right way, relaxing even further the constraints on their size.

39 One of these is the planned Laser Interferometer Space Antenna (LISA), a space-based version of LIGO comprising multiple spacecraft, separated by millions of kilometers, playing the role of LIGO's four-kilometer tubes. There are also other detectors that are playing a critical role in the search for gravitational waves, including the German-British detector GEO600, the French-Italian detector VIRGO, and the Japanese detector TAMA300.

40 Since teleportation starts with something here and seeks to make it appear at a distant location, in this section I will often speak as if particles have definite positions. To be more precise, I should always say, "starting with a particle that has a high likelihood of being located here" or "starting with a particle with a 99 percent chance of being located here," with similar language used where the particle is teleported, but for brevity's sake I will use the looser language.

41 For collections of particles—as opposed to individual particles—the quantum state also encodes the relationship of each particle in the collection to every other. So, by exactly reproducing the quantum state of the particles making up the DeLorean, we ensure that they all stand in the same relation to each other; the only change they experience is that their overall location would have been shifted from New York to London.

42 The fragility of the human body is another practical limitation: the acceleration required to reach such high speeds in a reasonable length of time is well beyond what the body can withstand. Note, too, that the slowing of time gives a strategy, in principle, for reaching distant locations in space. If a rocket were to leave earth and head for the Andromeda galaxy, traveling at 99.999999999999999 percent of light speed, we'd have to wait nearly 6 million years for it to return. But at that speed, time on the rocket slows down relative to time on earth so dramatically that upon returning the astronaut would have aged only